

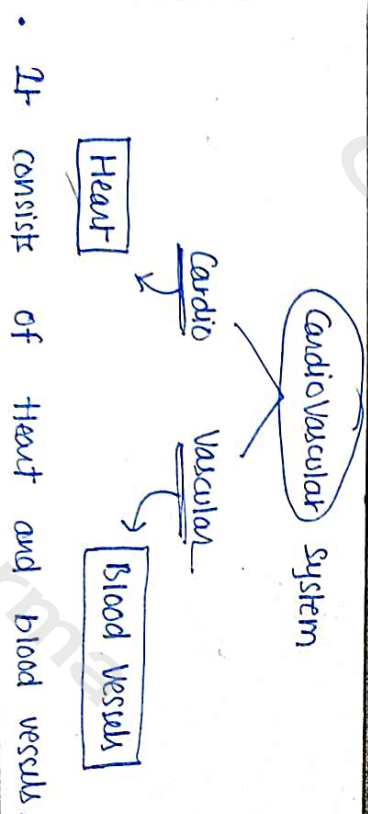
CAREWELL PHARMA - A FAMILY OF LEARNING

search CAREWELL PHARMA on Youtube, Watch free Video Lectures

UNIT-Ist PHARMACOLOGY-II

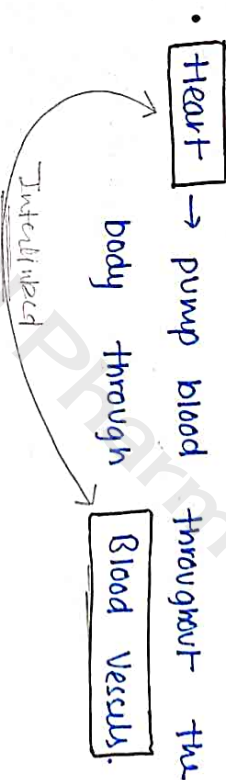
PHARMACOLOGY OF DRUGS ACTING ON CARDIOVASCULAR SYSTEM (CVS)

- Introduction to hemodynamic and electrophysiology of heart.
- Drugs used in congestive heart failure.
- Anti-hypertensive drugs
- Anti-anginal drugs
- Anti-arrhythmic drugs
- Anti-hyperlipidemic drugs.

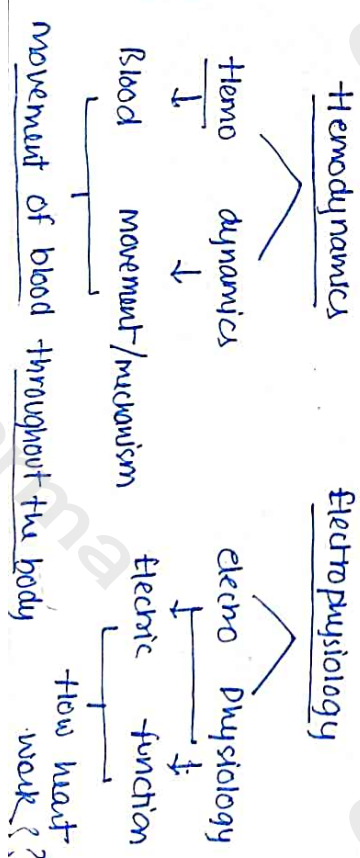


- It is also known as circulatory system.

whole body
transport blood to the
transport O_2 ,
nutrients to the body
essential for respiration



- Introduction to hemodynamic and electrophysiology of heart.



Website For Notes & More :- www.carewellpharma.in

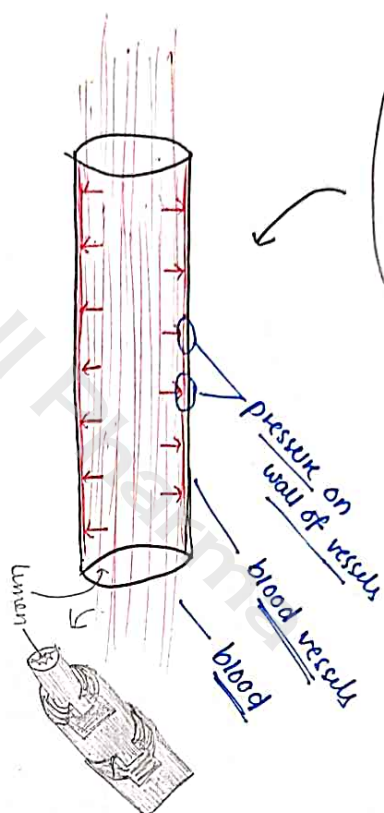
Visit Website and Download for free

CAREWELL PHARMA - A FAMILY OF LEARNING

search CAREWELL PHARMA on Youtube, Watch free Video Lectures

• Hemodynamics →

movement of blood in blood through blood vessels.



• Blood pressure → The pressure exerted by blood to the inner wall of blood vessels.

— It is depend upon the diameter of blood vessels.



— less diameter
— contraction (B.P ↑)



— more diameter
— dilation (B.P ↓)

— B.P. →

SYSTOLIC

(mmHg)

DIASTOLIC

Normal → < 120

< 80

Elevated → 120 - 129

80

High BP ↑ → 130 - 139 (stage 1)

8 - 89

High BP ↑ → 140 or higher (stage 2)

90 or higher

Hypertensive → 180 or more ↑ Crisis

Higher than 120 ↑

Website For Notes & More :- www.carewellpharma.in

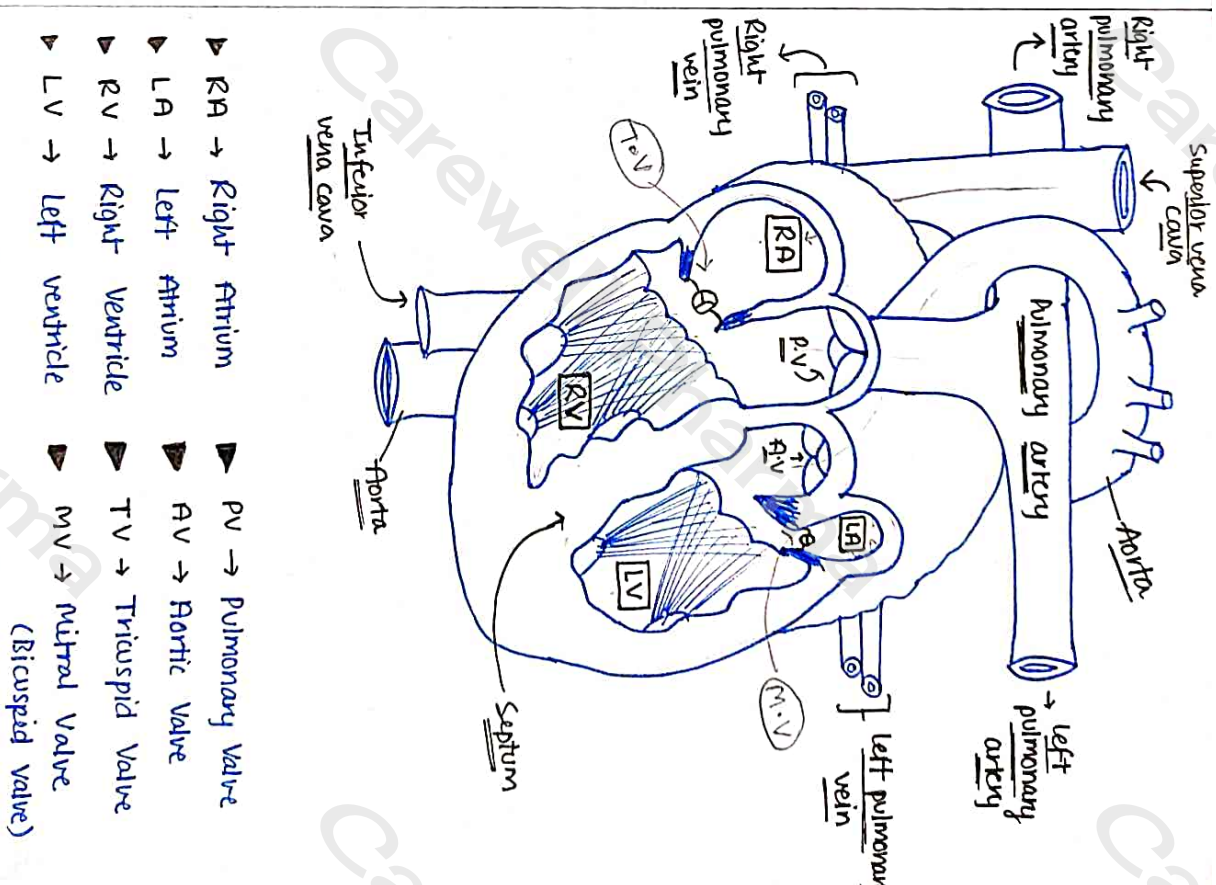
Visit Website and Download for free

CAREWELL PHARMA - A FAMILY OF LEARNING

search CAREWELL PHARMA on Youtube, Watch free Video Lectures

Heart

- It is a roughly cone-shaped hollow muscular organ. It is about 10cm long and it is about the owner's fist.
- Average weight of heart →
Male $\Rightarrow$ 300gm # female $\Rightarrow$ 250 gm
- Location →
lie in the thoracic cavity in the space b/w the lungs. (two third part of heart is on left side).
- layers of heart →
 - 1) epicardium → the outer layer
 - 2) Myocardium → middle layer (muscular layer)
responsible for pumping
 - 3) endocardium → the inner layer

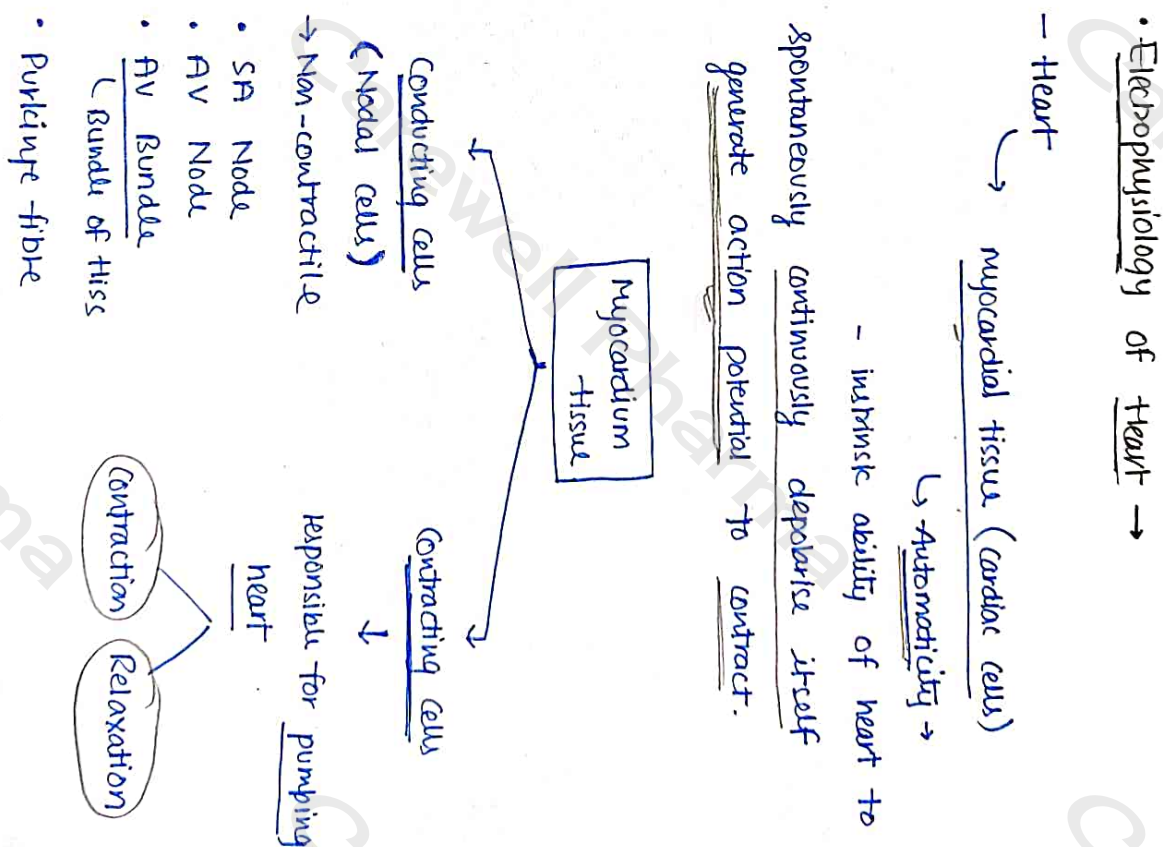
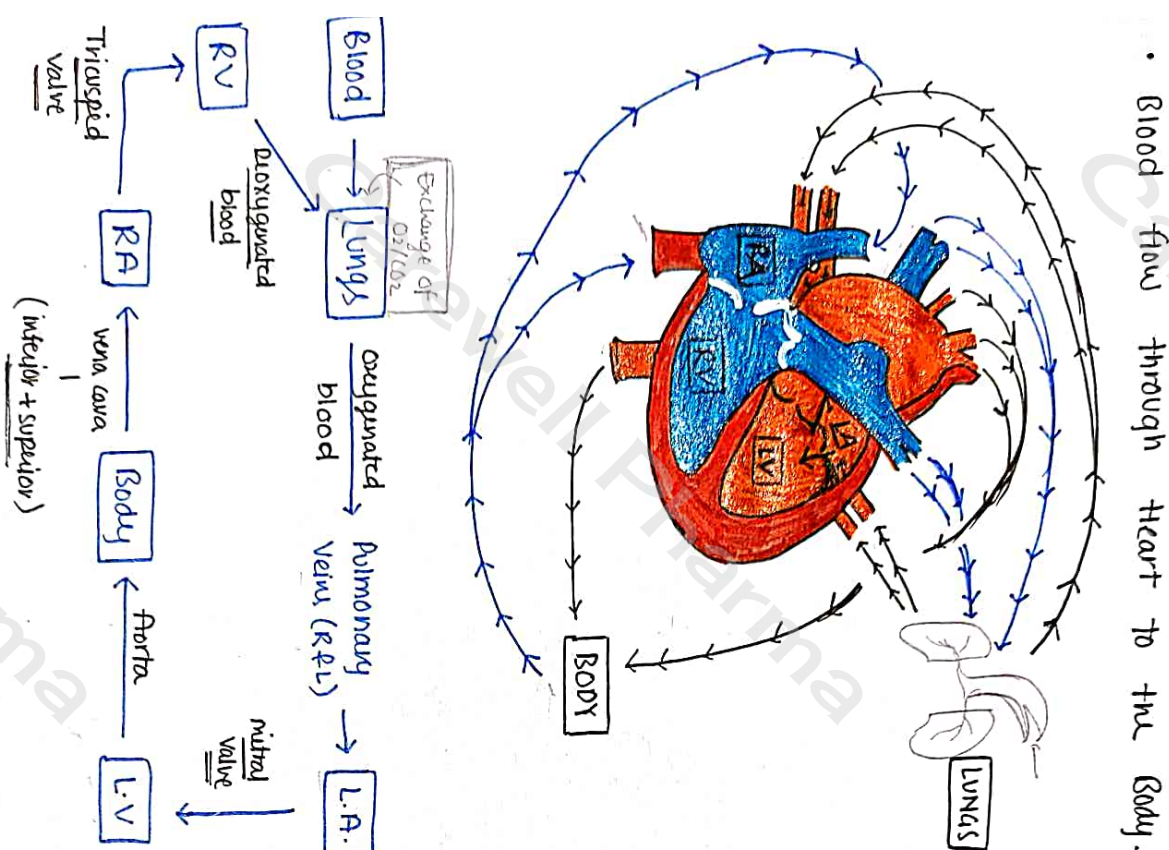


Website For Notes & More :- www.carewellpharma.in

Visit Website and Download for free

CAREWELL PHARMA - A FAMILY OF LEARNING

search CAREWELL PHARMA on Youtube, Watch free Video Lectures



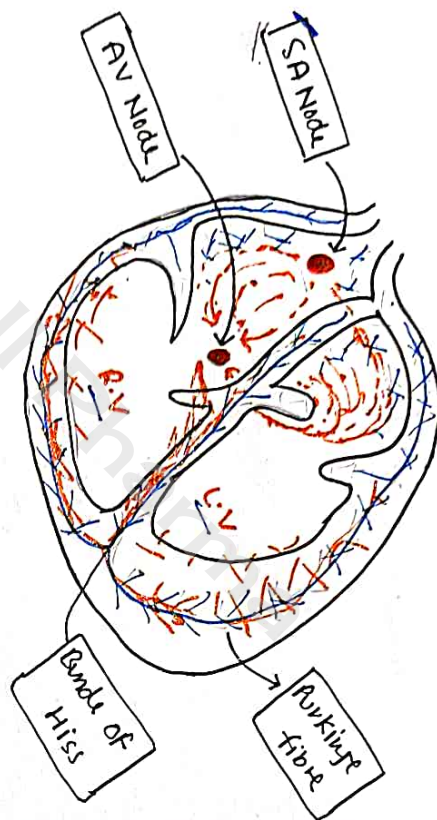
Website For Notes & More :- www.carewellpharma.in

Visit Website and Download for free

CAREWELL PHARMA - A FAMILY OF LEARNING

search CAREWELL PHARMA on Youtube, Watch free Video Lectures

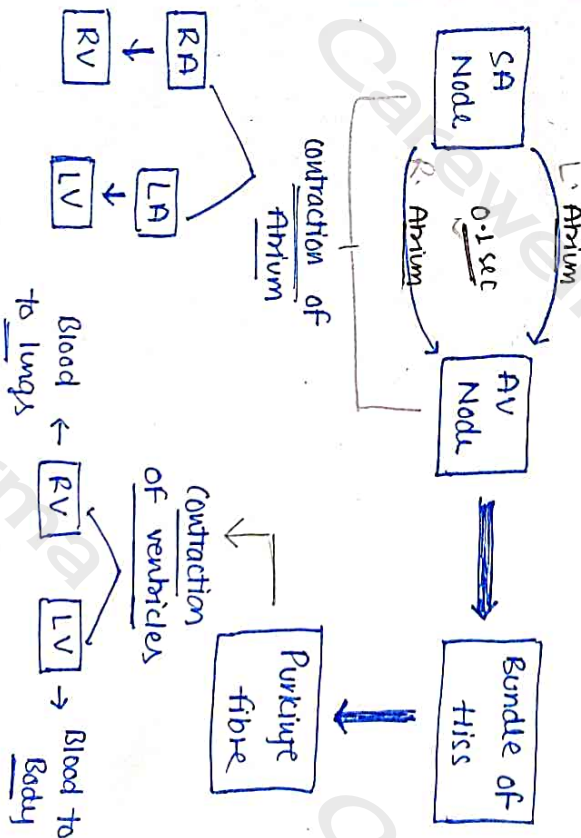
• Conducting system of heart



- SA node → Sinatrial Node
 - pacemaker of the heart
 - generate action potential (electric impulse)
 - located in the wall of right atrium near the superior vena cava.
- AV node → Atrioventricular Node
 - 2nd pacemaker
 - located in the right side of atrial wall just below the opening of superior vena cava.

• Bundle of His (AV bundle) →

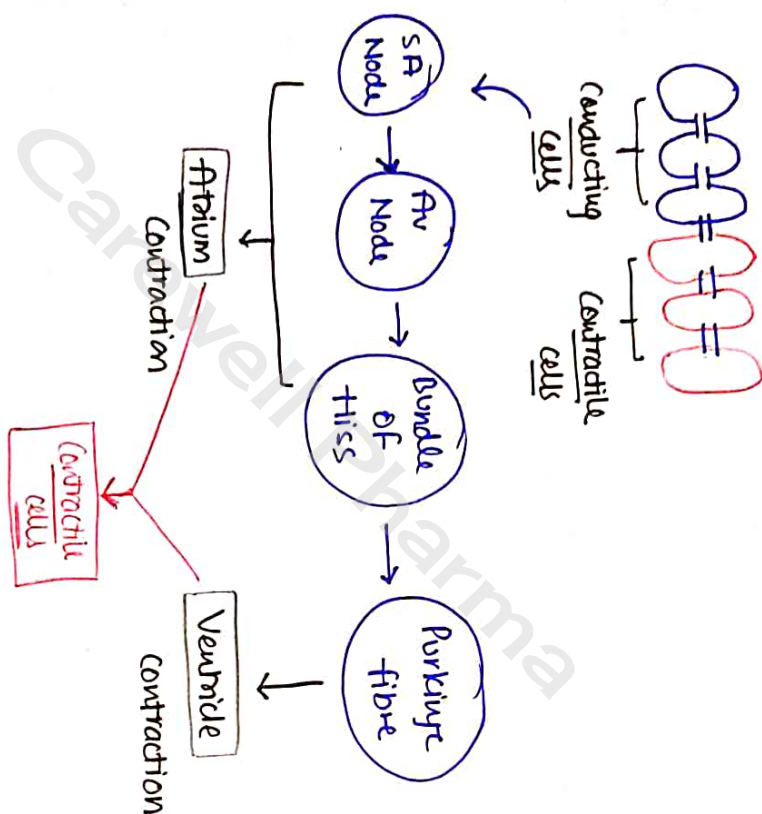
- located b/w the atria & ventricular.
- electric impulse travels from the AV node to the bundle branches
- Purkinje fibres →
 - located at the end of the AV bundle at the base of heart
 - responsible for contraction of ventricles & papillary muscle tips.



CAREWELL PHARMA - A FAMILY OF LEARNING

search CAREWELL PHARMA on Youtube, Watch free Video Lectures

• How it generate impulse →



1) Conducting cells →

Those cells which are responsible for generate current.

Resting Membrane potential @ -60 mV

funny Sodium channel → slowly Na⁺ influx

Ca²⁺ influx → Open T-type Ca²⁺ channel → change membrane potential (-60mV → -55mV)

change (-55mV → -40mV) → Open L-type Ca²⁺ channel → strong Ca²⁺ influx

threshold potential

+140mV

(-40mV → +40mV) depolarisation

K⁺ channel open → K⁺ outflux

Repolarisation

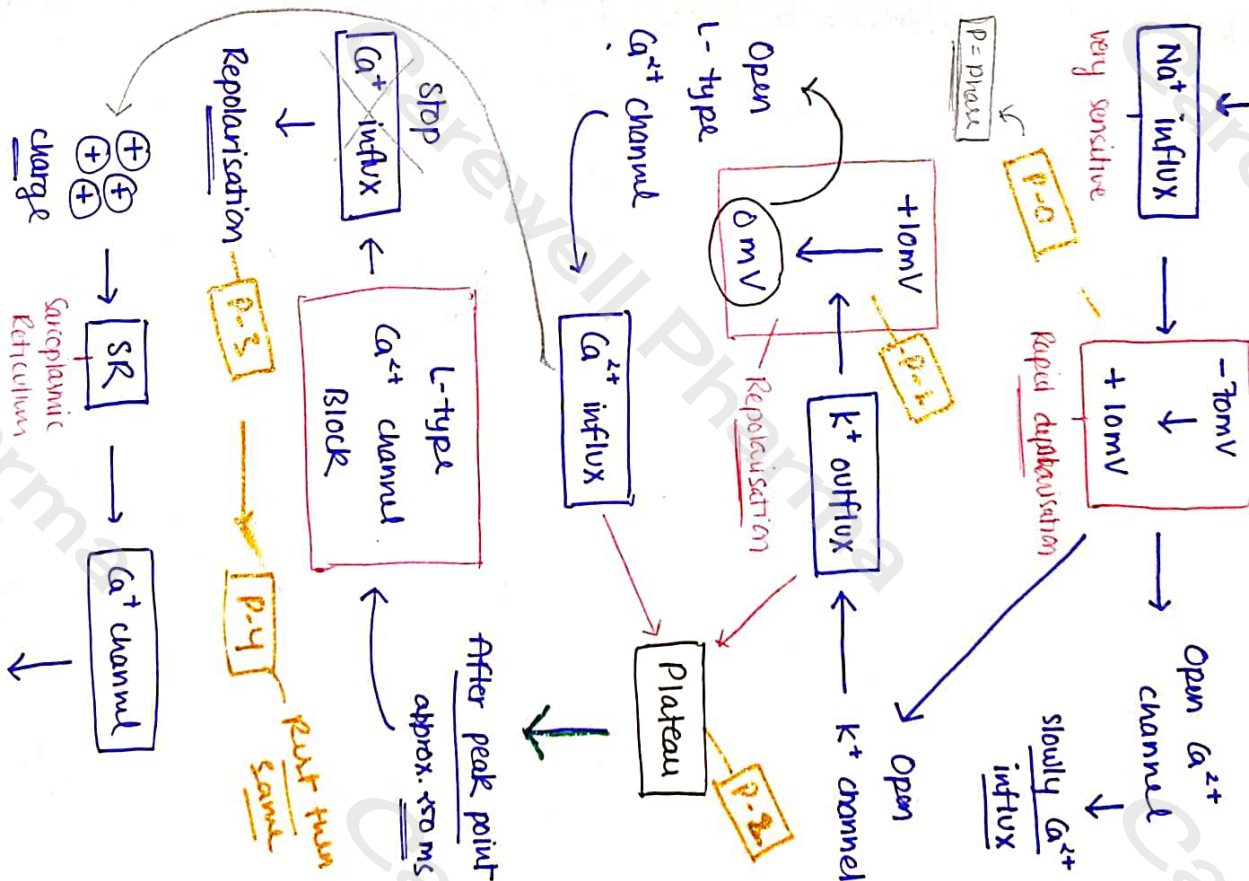
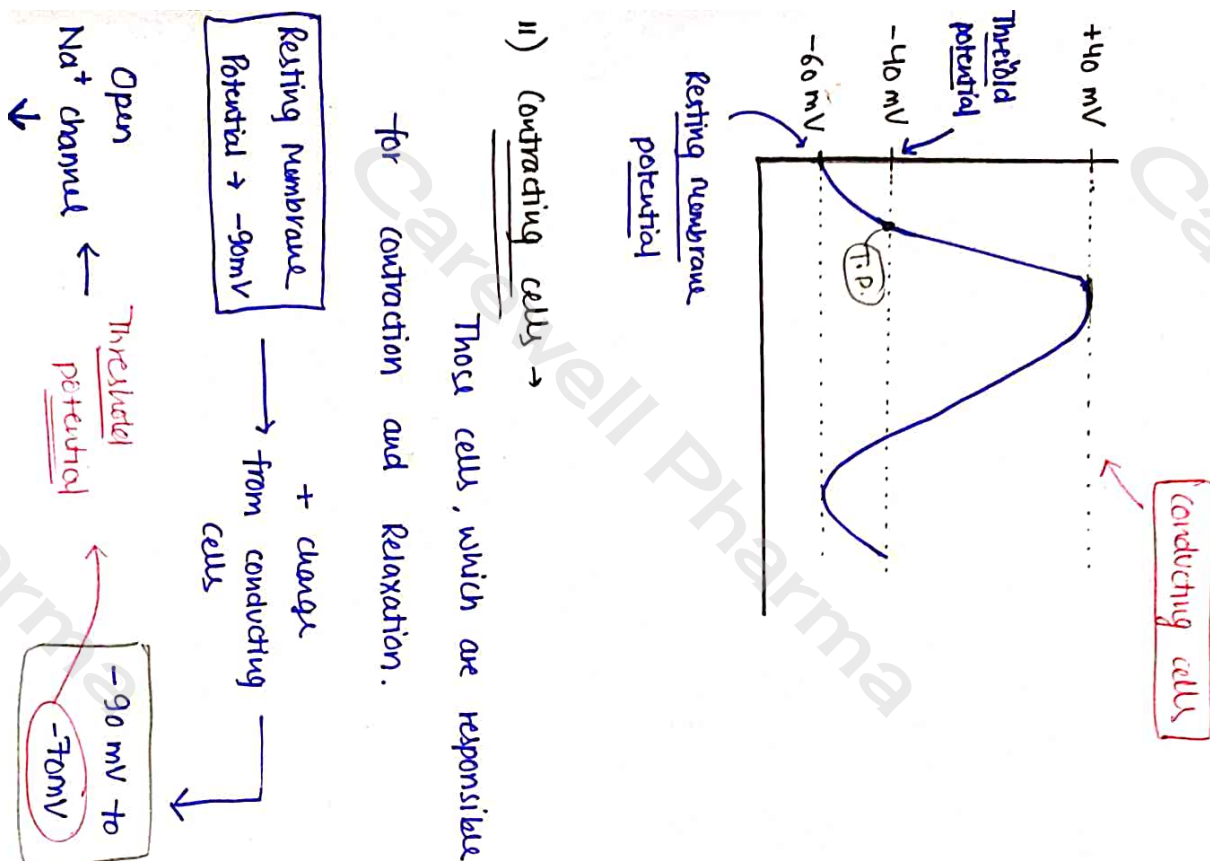
pass into contractile cells

Website For Notes & More :- www.carewellpharma.in

Visit Website and Download for free

CAREWELL PHARMA - A FAMILY OF LEARNING

search CAREWELL PHARMA on Youtube, Watch free Video Lectures

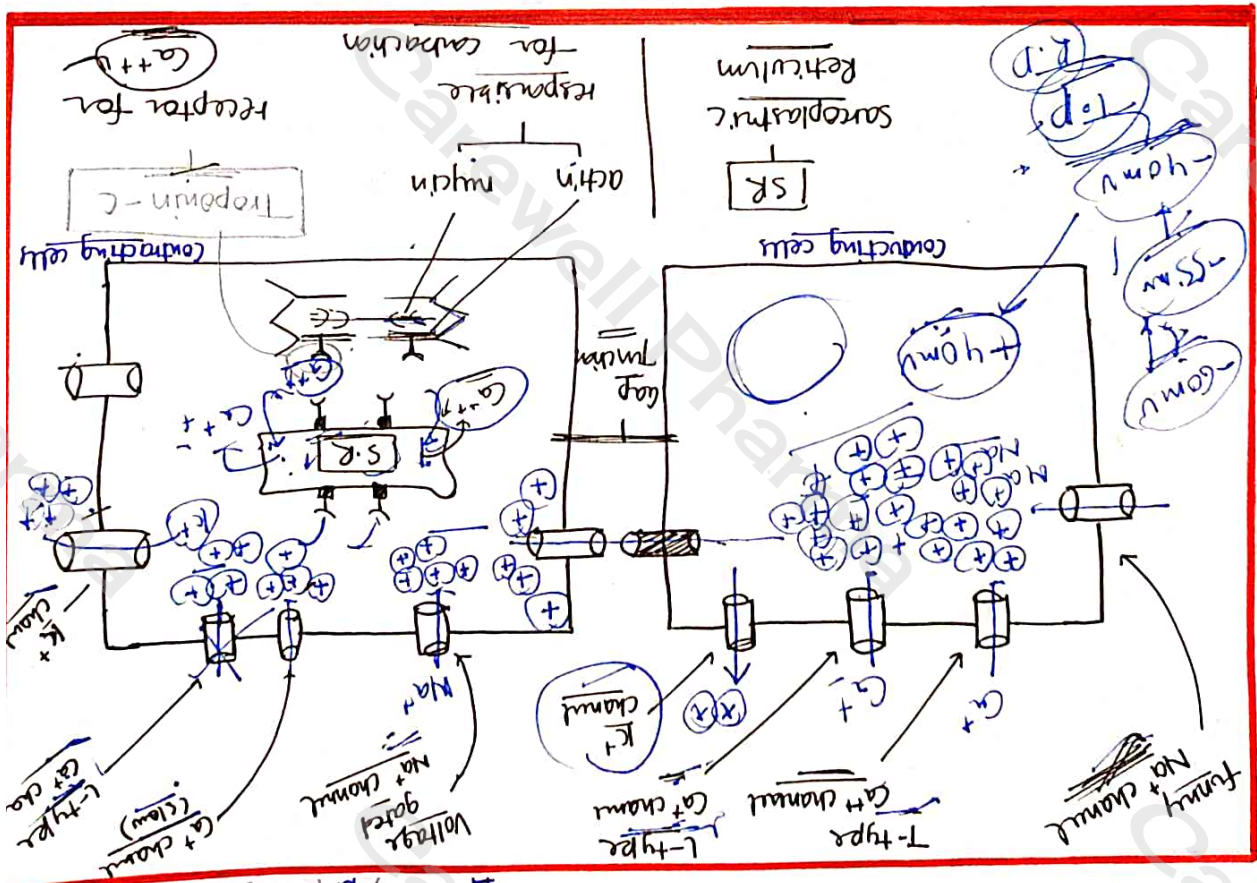
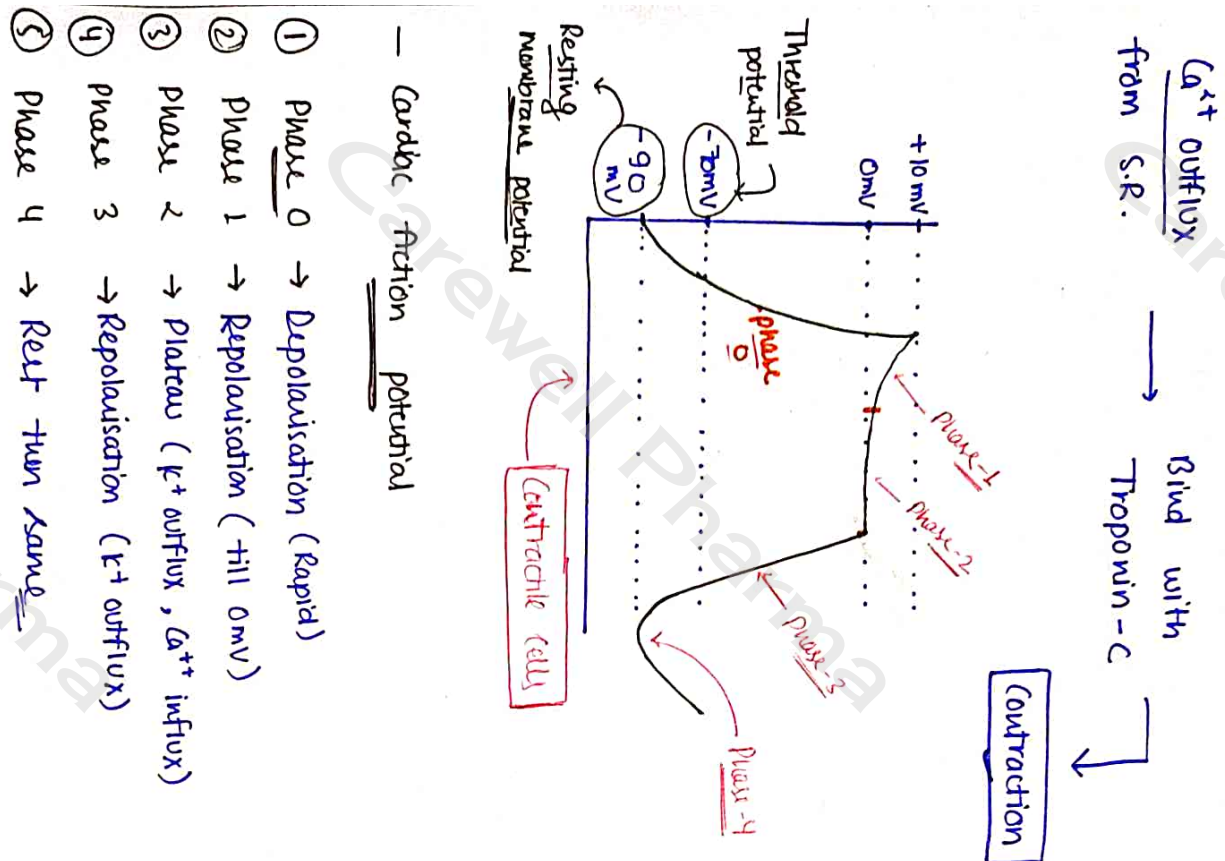


Website For Notes & More :- www.carewellpharma.in

Visit Website and Download for free

CAREWELL PHARMA - A FAMILY OF LEARNING

search CAREWELL PHARMA on Youtube, Watch free Video Lectures

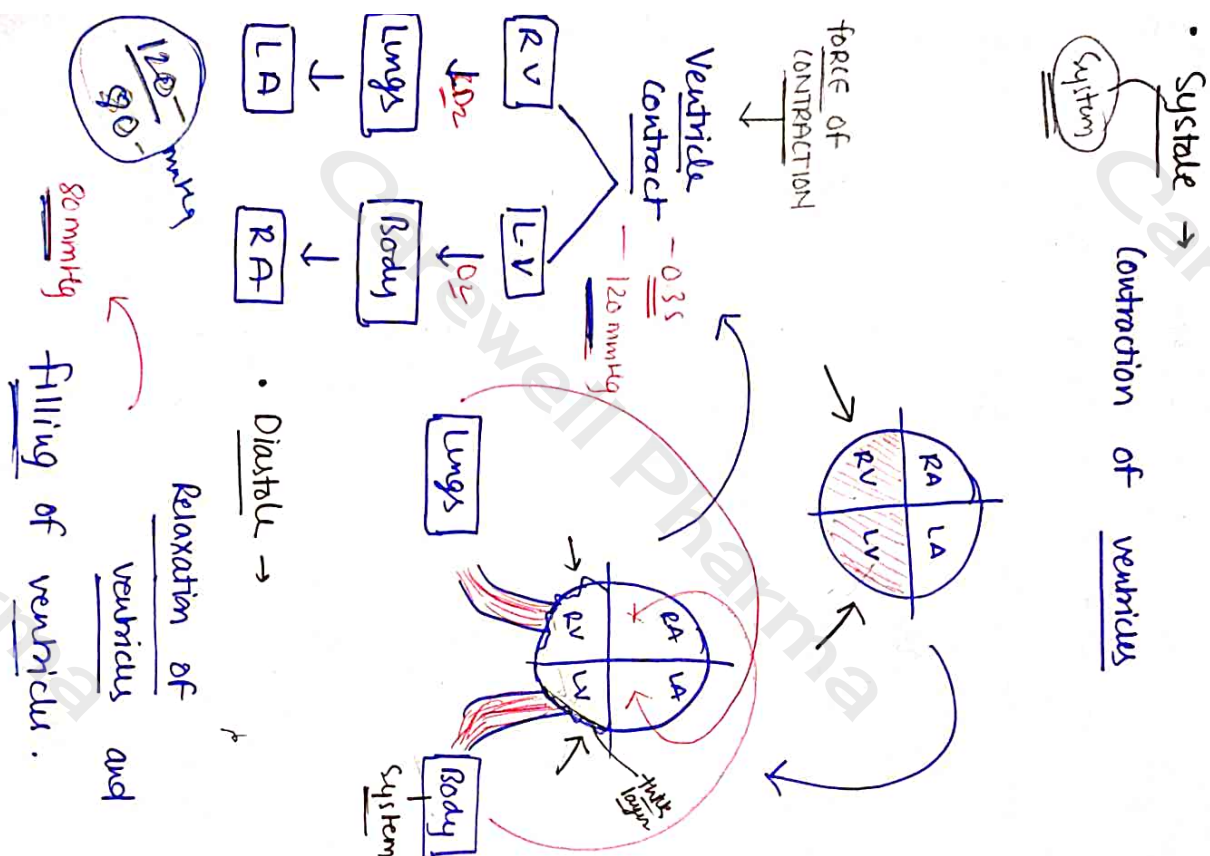


Website For Notes & More :- www.carewellpharma.in

Visit Website and Download for free

CAREWELL PHARMA - A FAMILY OF LEARNING

search CAREWELL PHARMA on Youtube, Watch free Video Lectures



• Cardiac output → It is defined as the amount of blood pumped out of each ventricle per minutes.

$$\begin{aligned} \text{C.O.} &= \text{Stroke volume} \times \text{Heart Rate} \\ &= \text{ml/beat} \times \text{beat/min} \\ &= \text{ml/min} \end{aligned}$$

- Stroke volume → volume of blood pumped out of each ventricle per beat.
- Average S.V. ⇒ 70-80 ml/beat
- Heart Rate → the speed at which heart beats.
- Average heart rate ⇒ 70-80 beat/min
- Average cardiac output → 5000 ml/min.
- A healthy heart with a normal cardiac output pumps about 5-6 litre of blood every minutes. (Rest)

Website For Notes & More :- www.carewellpharma.in

Visit Website and Download for free